

Effectiveness of Video Self-Modeling to Promote Social Initiations by 3-Year-Olds With Autism Spectrum Disorders

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Abstract

Video Self-Modeling (VSM) provides individuals the opportunity to view themselves performing a task beyond their present functioning level through the careful editing of videos. In this study, a single-case multiple-baseline design was used to determine whether VSM would facilitate social initiations across three young children ($M = 3$ years 10 months) on the autism spectrum. Although VSM has been found to enhance skills in younger children with autism spectrum disorders, no changes in behavior were noted for the participants in this study. The relationship between age and VSM efficacy is discussed along with other factors that may influence VSM outcomes with young children.

Keywords

video self-modeling, autism spectrum disorders, initiations, young children

Video modeling is gaining respect as a treatment option for children with autism and other developmental disabilities (Ayres & Langone, 2005; Delano, 2007; Hitchcock, Dowrick, & Prater, 2003; McCoy & Hermansen, 2007). These reviews of literature and one meta-analysis (Bellini & Akullian, 2007) have revealed a growing body of research supporting video modeling use across a range of behaviors and ages. Several reasons are given for video modeling's success, including the tendency for children with autism to be visual learners, the minimal distractions offered by the medium, and the lack of demands for children to socially interact with the videos (Buggy, 2007).

A substantial proportion of video-modeling research with children with autism involves video self-modeling (VSM) in which children view edited videos of themselves performing appropriately or at an advanced level. Dowrick (1983) coined the term *feedforward* for the process of obtaining information about potential, positive future functioning of an observer. In addition to the advantages listed for video modeling, VSM takes advantage of the obvious similarities of the model and the viewer which Bandura (2001) found to be a key component to the model's efficacy. Bandura found that the best models were those most similar to the observer in all characteristics, including appearance, age, and ability. Likewise, Bandura studied the importance of self-efficacy or the belief that one can be successful at a task. He found a high correlation between self-efficacy and success. VSM may provide a significant boost to one's self-efficacy by providing an individual with visual evidence of successful performance.

The number of studies using VSM in all forms with persons with autism is small, although growing. Ayres and Langone (2005) reviewed all forms of video-modeling methods, including adult, peer, and self, used with persons with autism and found that authors in 4 of the 14 articles addressed self-as-model. Likewise, Delano (2007) found that 5 of 19 studies on video modeling involved VSM. Bellini and Akullian (2007) took the review of literature one step further by conducting a meta-analysis of studies using video modeling with children with autism spectrum disorders (ASD). They found 23 such studies, including 8 that evaluated VSM use with persons with autism. Findings across studies can be interpreted to conclude that acquisition of new skills was greatly facilitated, the generalization of taught skills was occurring, and the maintenance of skills was strong for all forms of video modeling. Bellini and Akullian concluded that all forms of video modeling, including VSM, met the Council for Exceptional Children's guidelines (Horner et al., 2005) for research-based practices.

As one might expect, the number of studies in which preschool children with autism have been participants is extremely limited. In fact, only four were identified in the search of the literature for the present study with a total of

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11 participants. Wert and Neisworth (2003) used VSM to train spontaneous requesting in 4 preschoolers with autism. The children were trained to request items via a discrete trial method; however, the resulting requests were rote and there was no generalization to spontaneous requesting without prompting. A video was created with the prompts edited out and only the best examples included. Results, collected using a multiple-baseline design across individuals, indicated that all 4 participants showed substantial gains. The gains in mean production of spontaneous requests ranged from 800% to 1,200%.

Buggey (2005) worked with a 4-year-old boy to try to reduce aggressive (pushing) behavior and to increase responses to questions and verbal initiations. A multiple-baseline design across three behaviors was used for evaluation purposes. A decrease in pushing to almost zero occurred and gains were reported for both verbal behaviors. The video addressing pushing depicted replacement behaviors such as touching appropriately and compliance with teacher requests.

Bellini, Akullian, and Hopf (2007) examined the effects of VSM on the social engagement of two 4-year-olds with ASD. The authors stressed the importance of obtaining data in natural environments and accordingly recorded percentage of time engaged with peers during normal play times in the classroom. The videos were constructed using footage from brief interactive periods captured in the classroom. Following the children's viewing of their videos, the percentage of time spent engaged in play with peers increased from 3% to 43% and 6% to 24%, respectively. As in other studies, a strong maintenance effect was seen following withdrawal of VSM.

Finally, Buggey, Hoomes, Sherberger, and Williams (2011) worked with four children, three of whom just turned 4 and one who was 3 years 10 months at the outset of the study. All were diagnosed with moderate to severe autism. The targeted behaviors were social initiations on the playground. A multiple-baseline design across children was used to evaluate results. One or two peers along with the participant were taken to the playground and prompted to interact while being filmed. Some scenes were staged such as taking the participant and peer to the top of a slide, getting them to hold hands, then gently pushing them off. The 2- to 3-min videos were edited so it appeared that the children were interacting frequently and appropriately with their peers. The children watched the videos in their classrooms on arrival at school for eight successive mornings. Buggey et al. reported increased initiations and verbalizations as well as positive changes in several behaviors not specifically targeted in the videos for three of the four participants. The one child exhibiting no changes in behavior was the youngest participant.

Overall, researchers have reported positive results for VSM. The one exception to this trend was a study with preschoolers

(Clark et al., 1993). Although the six participants were categorized as having oppositional-defiant disorder rather than autism, their ages (i.e., between 3 and 5 years) and some of their behaviors were similar to those in the studies of preschoolers with autism. A complex reversal design was used to compare VSM with peer-modeling. No change was evident in the reduction of aggressive behaviors with either method.

With the exception of the Clark et al. (1993) study, the results obtained with preschool participants have been consistently positive, yet research in this area is so limited that it is difficult to make claims about who might benefit from VSM and what factors might contribute to success or failure. It is logical to assume that VSM must have limits based on developmental skills related to children's ability to model or self-recognize. Other factors such as temperament, complexity of behaviors being addressed, attention span, or some combination of these factors also may contribute. The selection of appropriate behaviors that can be effectively communicated via the video medium for very young children also is problematic.

The purpose of the present study was to investigate whether or not VSM could be used to increase the number of social initiations of 3-year-olds with ASD during playground activities. It was designed to be a replication of the work of Buggey et al. (2009) with the difference being the age of the participants. The children in the present study were an average of 4 months younger than the children who participated in the earlier study.

Method

Participants

Much of the diagnostic information on the children was obtained very close to their third birthdays as they prepared to transition from early intervention to public school preschool services. Certified school psychologists from the local public school system carried out assessments. Parents and teachers assessed the children on several measures. Assessment results for each child are presented in Table 1.

Child 1. Interestingly, for Bob, his mother and teacher completed *The Gilliam Autism Rating Scale* (GARS-2; Gilliam, 2006), with somewhat conflicting results. The mother's percentile score indicated a below-average probability of autism while the teacher's score was indicative of average probability of autism. The same persons completed the second edition of the *Adaptive Behavior Assessment System* (ABAS-II; Harrison & Oakland, 2003) and obtained closely aligned scores in the domains of General Adaptive, Conceptual, Social, and Practical. Percentile scores across subscales and evaluators ranged from 1.00 to 8.00, indicating significant delays in all areas. The assessments were carried out when Bob was 2 years 10 months, and the

Table 1. Participant Evaluation Results

Characteristic/assessment instrument	Bob	Tevon	Josh
Age	3 years 8 months	4 years 1 months	3 years 8 months
Diagnosis	PDD-NOS	DD changed to PDD-NOS	PDD-NOS
ABAS-II parent/teacher-scaled scores: total score ^a	36	39	47/59
BSITD-III composite score/percentile rank	55/0.1	No score	55/0.1
GARS-2 standard score/percentile			
Parent	87/19	91/27	83/13
Teacher	91/27	102/55	89/23
CARS	N/A	39.5—"severely autistic"	45—"severely autistic"

Note: PDD-NOS = pervasive developmental disorder—not otherwise specified; DD = developmental disorder; ABAS-II = *Adaptive Behavior Assessment System* (2nd ed.; Harrison & Oakland, 2003); BSITD-III = *Bayley Scales of Infant and Toddler Development* (3rd ed.; Bayley, 2005); GARS-2 = *Gilliam Autism Rating Scales* (2nd ed.; Gilliam, 2006); CARS = *Childhood Autism Rating Scale* (Schopler, Reichier, & Roehen Renner, 1988).

^aIndividual scores represent teacher evaluations.

conclusion was that he exhibited the behaviors consistent with a diagnosis of autism.

Observations carried out on the playground by the author revealed a child engaged in solitary play. He played on equipment, especially the swings, slides, and climbing equipment. Bob would "hover" around the swing area waiting for an opening. He often climbed to a height of up to 3 feet and then jumped to the ground. When other children approached him or there was loud noise in his vicinity, he raised his elbows to face level and pressed in on his cheeks with his fists. He did not show interest in the play of others. He did engage in self-talk using simple sentences but was not observed communicating to others.

Child 2. When Tevon was 2 years 8 months of age, an administration of the third edition of the *Bayley Scales of Infant and Toddler Development* (Bayley, 2005) was attempted but a "standardized score could not be obtained." On the ABAS-II, completed by his mother and teacher, Tevon obtained percentile rank scores below 9.00 in the areas of Conceptual, Social, and Practical skills, with the teacher rating him slightly higher than did the mother. All ratings were indicative of significant developmental delay. The scores of mother and teacher on the GARS indicated a high likelihood of autism. Tevon's teacher completed the *Childhood Autism Rating Scale* (CARS; Schopler, Reichier, & Roehen Renner, 1988) and obtained a Total Score of 39.5, indicating severe autism.

Tevon was observed making direct eye contact with peers especially with those with physical disabilities or younger children on the playground. He sometimes placed himself nose to nose with the peer while patting him or her gently and verbalizing in soothing tones. He also held hands with a peer on the way to the playground. On the playground, he exhibited a high degree of mobility. He was on the move often and approached others, although his attention seemed very fleeting in these interactions, rarely lasting more than 5 s.

Child 3. A developmental assessment for Josh was conducted just prior to his third birthday. His lack of cooperation during *Bayley* administration may have made the score a low estimate. Assessment with the ABAS-II yielded percentile ranks of 0.2 in Conceptual, 0.6 in Social, and 0.9 in Practical (functional skills). All ratings were indicative of significant developmental delay.

On the playground, Josh typically was in constant motion following a set path that circled the swings and two of the large pieces of play equipment. Occasionally, he circled a small storage building that resembled a garage, stopping and looking at each window. He liked to swing and approached adults and took them by the hand when he saw a vacant seat. Josh was easy to track on the playground because he often made a protracted "eee" sound as he moved. He rarely interacted with others, although occasionally he lightly hit peers as he passed them on the long ramps leading up to the slides. These hits accounted for all of his interactions during baseline.

Eligibility. To determine whether the children could self-recognize, the same method was used as in Buggey et al. (2009). Children were allowed to see themselves by turning a relatively large view screen on a camcorder so that it was facing them. Their reactions to their images were noted. All of the children attended well to their images. They stopped what they were doing and maintained eye contact with the screen for more than 30 s. Bob and Tevon made faces or moved around while maintaining eye contact. This display of cause-effect behavior aimed at the camera was considered a good indicator of self-recognition. Josh smiled at the screen but did not make noticeable movements to affect the camera image. Although the duration of his viewing was prolonged more than 30 s, it was less clear if he self-recognized. Josh's reaction to the windows of the small building he circled on the playground indicated that self-recognition was likely.

Prior to the outset of the study, approval was obtained from the university's Internal Review Board, and the

Research Committee at the institute where the study was carried out. Informed consent was obtained from the parents of the participants and amendments were added to the children's Individual Education Plan (IEP) regarding the treatment.

Setting

Three students who were diagnosed with an ASD and who attended a private inclusive preschool in a small Southeastern city were the focus of this study. The preschool was part of a larger institute that featured a Developmental Pediatric Clinic and a Family and Child Research Center; the preschool also served as an observation site for several local colleges. The common presence of adult observers made the children somewhat immune to observer effects. The preschool had an equal number of children with and without disabilities and this ratio was reflected in each classroom. The classrooms were arranged by age ranging from 6 weeks to 1 year in the youngest classroom to children 4 to 5-plus years in the oldest. The three participants were in separate classrooms for children who were approximately 3 to 4 years of age. The classrooms each had 12 children, 6 of whom were typically developing and 6 who had disabilities. Each class was staffed by a lead teacher, two aides, and an additional aide assigned to each study participant. Children received a full range of support services, including speech, physical, and occupational therapies that were delivered within the classrooms when possible. The children in the present study each participated in occupational and speech/language therapies. The classrooms were typical with mobile low storage shelving serving as dividers for centers. Centers varied in content and revolved around a theme that was initiated in classrooms. The playground area where filming and data collection were carried out was approximately 10,000 square feet with modern, multicolored equipment. Three large structures with elements for climbing, sliding, and physical manipulation dominated the playground with adjoining swing and sand areas. The three main structures were designed for children of different motor development levels and were wheelchair accessible.

Materials/Equipment

Video was taken with a Sony© Handycam DCR-TRV22 mini DV camcorder and edited on an Apple© MacBook© laptop computer using the built-in iMovie© software program. The newer iMovie© program allows for cropping film clips which allowed the team to remove adults from the movie when they came into camera range and to zoom in on the children without losing too much quality. Other features of iMovie© that were used in the production of the self-modeling video were titles, transitions between clips, and audio insertions, including clapping, music, and

voice-overs. Sound effects and ambient music come with the software, so applying these to the video was simply a matter of clicking and dragging the content. (The same formatting is used for the PC software counterpart, Movie Maker©.)

The filming sessions on the playground utilized the material and equipment that were always present. Along with the larger playground equipment mentioned in the setting section, balls of various sizes, digging toys for sand play, and tricycles and small enclosed plastic cars were used in the videos and were available for use during recess.

Design and Analysis

The study was conducted using a single-case, multiple-baseline design across three children consisting of baseline, intervention, and maintenance phases. The study consisted of two stages. The first began in January and concluded in June. These same phases were repeated in a follow-up stage of the study that was conducted the subsequent fall, at which time the children's ages closely aligned with those in the study conducted by Buggey et al. (2009). Two of the children changed classrooms for the second stage of the study conducted in the fall, but all other elements remained the same. Two methods were used to analyze data. The first was visual inspection of graphs, and the second was differences in means across study phases.

Dependent Variables

The number of physical and vocal social initiations with peers during playground time was selected as the dependent variable. The definition for the dependent variable was identical to that used in the study by Buggey et al. (2009). *Social initiation* was divided into two subcategories: *physical* and *vocal*. Physical initiation was defined as moving into proximity of one or more peers with resulting *engagement*. Engagement was defined as either attending to the peer(s) or the activity of the peer(s) for more than 5 s while remaining at arm's length or making direct physical contact with peers. Vocal initiation was defined in terms of use of words (verbal) or nonverbal vocalizations such as laughing, crying, and other sounds that were directed at a peer. If a vocal initiation was observed, then the simultaneous physical movement into proximity of the peer was not recorded. Thus, each initiation was only scored once and vocalization was given priority over physical approach. The definition of *Initiation* was defined so that repetitive activities (such as rolling a ball back and forth) would only be tallied once. Thus, once a physical initiation was carried out, a child needed to disengage completely from the peer for 10 s before another initiation would be recorded. Vocal events were similarly not scored except for the first event; however, verbalizations (actual words or phrases) separated by

at least 5 s and not identical to the initial verbalization were recorded as distinct data points.

Observation and Data Collection

Two observers collected data: the author who is a university professor and a graduate assistant in the early childhood education master's degree program. Observers had data collection forms on clipboards and moved as close to participants as possible so that verbalizations could be heard. The observation form is available on request; however, it had a simple design with columns for type of initiation (physical or vocal), time of initiation, and comments. The comments column was used for recording information about the behavior, including very brief descriptions, relevant antecedents, sites, persons involved, and other environmental factors. The author spent considerable time on the playground over the previous 2 years carrying out other observations so most of the children were familiar with him. The other observer spent a month on the playground prior to collecting baseline data. Much of the initial month was spent observing and establishing interobserver agreement. In a further effort to blend in on the playground, the observers interacted with children between observations.

An event-recording approach (Kennedy, 2005) was used for collection of data. The observers conducted daily, 15-min observations of participants during playground time. Each of the children was observed at least 1 time per day. A chart was kept of the number of observations per child and the observers did not observe the same child simultaneously unless interobserver agreement was being carried out. The order of viewing the children was dependent on when the classrooms came out for their playground time. Playground time was set for 9:30 each morning, but there was considerable variance.

Interobserver Agreement

One month prior to beginning the baseline phase of the study, the interobserver agreement process began. The observers collected data on the children during playground times. Because the children with an ASD rarely interacted, many of the observation sessions involved typically developing peers or other children with developmental disabilities. The observers then met to review their findings and discuss differences of opinion. The observers continued to collect and compare data until 85% agreement was achieved over five successive observations. In order for there to be agreement, both observers had to have a similar description of the behavior, the name or description of the child interacted with, the exact site, and a recorded time that was within 2 min of the other observation. Two minutes was chosen to account for the possible variations of reading seconds from watches, which were synchronized to minutes,

but not seconds. There also was a need to maintain visual contact when behavior occurred and it was found that recording events to the minute rather than second was practical. Requiring a triangulation of time, event description, and peer(s) involved offset any issues related to times not being perfectly aligned. A frequency-ratio approach was used for calculation of total agreement. This involved taking the sum of occurrences recorded by each observer and dividing the smaller total by the larger total and multiplying by 100% (Kennedy, 2005). Identical data collection procedures were used in the baseline, intervention, maintenance, and follow-up phases of the study. The same two observers participated throughout.

Thirty percent of the observations carried out in the initial baseline, intervention, and maintenance phases involved interobserver agreement along with 25% of the observations in the follow-up stage. Interobserver agreement ranged from 88% to 95% across the three phases in the first stage of the study and 89% to 97% in the second stage of the study.

Independent Variable

The independent variable was designed to be identical to the one used by Buggey et al. (2009). Videos were created depicting participants socially interacting with peers with an emphasis on initiations. The children in this study rarely initiated socially in an appropriate manner and had limitations in their ability to follow directions. Thus, video clips had to be compiled that gave the impression of social initiations.

Video footage was obtained by prompting peers to interact with the participants on the playground. The children with an ASD were taken to the playground with one or two peers in the afternoon when no other children were present. The peers were prompted to get the child with an ASD to interact with them (e.g., "Take Tevon over to the slide"; "Give Josh the shovel"). The software involved in the editing process allowed the researchers to highlight and emphasize appropriate behaviors. For example, a peer would be prompted to hold the hand of the participant as they moved across the playground. The child with an ASD may only have sustained the hand-holding for a few seconds; however, the few seconds could be captured and a freeze frame created just prior to the release of hands. Freeze frames can be made to be any length allowing for an extended scene of holding hands. All of the children enjoyed playing on swings. There was a three-person tire swing and conventional swings which were used in all of the videos. The author selected clips from the tire swing in which the participants were laughing, looking at peers, or performing a behavior that approximated social initiation such as leaning toward a peer. The mean amount of raw footage taken of each child was 15 min. Informed consent for participation of the peers was obtained from their parents.

The author created the videos for the three participants using iMovie[®] software on an Apple MacBook[®] laptop. Clips were edited and arranged to make it appear as if the child with an ASD was initiating interactions and playing with peers in an appropriate and fun manner. A very flattering still-frame image of the child was selected and placed at the beginning of the movie and the target behavior was verbally labeled (e.g., “This is _____’s Movie! Let’s watch _____ playing nicely with his friends.”). Even though the children could not read, titles were placed on the still-frame to better resemble actual TV programs. Similarly, another flattering still-frame was placed at the end of the movie and “Good job _____ playing with your friends!” was recorded over it. An audio clip of children cheering and clapping followed this. Ambient background music was used to eliminate background noise and adult talk including prompts. When the child and peers were talking or laughing, the music was turned off and the actual sound was turned back up. iMovie[®] was used for editing because of its simplicity and ready availability to many therapists, teachers, and parents. Video editing, which traditionally has been a complicated process, has been made accessible to anyone with a computer.

The videos were very similar in content and format. Because each of the children liked to swing, they were shown on the three-person tire swing “having fun” with peers. Footage was chosen that best depicted interaction, and any indication of less desirable behavior was eliminated. For example, Tevon liked to role his eyes revealing only the whites while spinning. This was cut from the final video while scenes where he was leaning toward his peers were included. To depict the idea of initiating play, physical approach was used. Video showing the participants moving into proximity of peers (although in reality they passed right on by) was included. Then still-frame images were inserted that were the best approximation of those children playing; thus, the participants could see themselves walking up to a peer followed by approximately 5 s of a photo showing them together. All of the videos had short vignettes of the participants transitioning from the classroom to the playground as well as footage of the children with peers on the large pieces of equipment.

All of the videos were 2 to 3 min in length. Dowrick and Raeburn (1995) stated that the optimum length of VSM clips is 2.5 min with times greater than this not producing any differences in effect. A MacBook[®] laptop with the video ready for play was brought to each classroom for viewing and teachers were instructed how to operate the computer. Teachers were asked to show the video to the children on their arrival at school. Teachers monitored the children as they viewed their videos and reported that all three were able to attend to the screen for the entire session although some difficulty was experienced at first with two of the participants. When Bob was first presented the laptop

and video, he began to cry and became distressed. The teacher believed this was due to the fact that Bob was used to playing games on the computer and that the video was a change in routine. The teacher prepared a picture symbol that was added to his visual schedule depicting the video of himself (TV with stick figure) in contrast to the ordinary computer symbol. This seemed to settle him down and he exhibited no stress thereafter. Josh showed initial interest but did not sit through the entire video for the first two showings. He did so for the subsequent viewings. Tevon had no difficulty with the video and watched it enthusiastically from the outset.

Once the first video was created, and it was clear that the filming process had not changed baseline rates of behavior, Bob began to view his video. The viewing occurred approximately 1 hr prior to recess which was approximately arrival time at the preschool. Bob continued to view the video for 8 sessions and then it was withdrawn. Eight viewings was a rather arbitrary limit but was based on the research evidence that changes occurred in a short time after viewing, if it was to happen at all (Buggey, 2007; Dowrick, 1983). Tevon began viewing his video shortly after Bob’s was withdrawn. This pattern continued until all three students had viewed their video.

Similar procedures were followed during the follow-up stage of the study. The viewing of the videos was limited to 5 consecutive days. Again, this number of days was somewhat arbitrarily but based on the reports of VSM causing immediate results or no results (Buggey, 2007).

Treatment Fidelity

Teachers were contacted every other day via email or direct contact and asked how the viewing went that day. Specifically, they were asked if there were any problems with the technology, how the child responded to the video, and were there any changes in the routine of presentation. No data from direct observation were collected to document treatment fidelity.

Results

None of the participants in this study appeared to make gains in the frequency of their social initiations during either stage of the study based on visual inspection of the data. The results per data collection session from the first stage are shown on the graph in Figure 1 while Figure 2 depicts the results from Stage 2. A summary of the mean rates of initiations across phases of the initial study is illustrated in Table 2 followed by the means from the follow-up stage in Table 3.

Tevon was the only participant who exhibited any change in frequency of initiation. His rate of initiation more than doubled between baseline and intervention; however, there

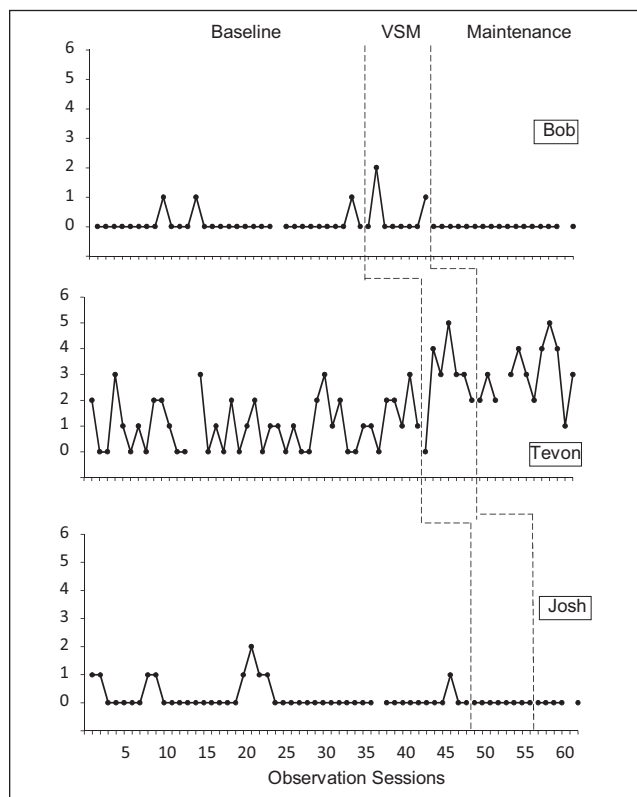


Figure 1. Numbers of Initiations per 15-Min Observation Sessions Among Study Participants: Stage I
Note: VSM = Video Self-Modeling.

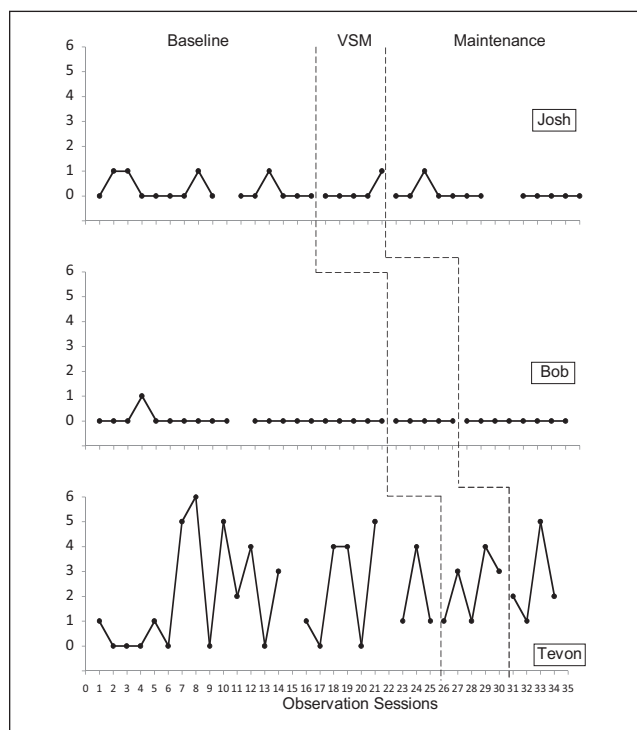


Figure 2. Numbers of Initiations per 15-Min Observation Sessions Among Study Participants: Stage 2
Note: VSM = Video Self-Modeling.

Table 2. Mean Numbers of Initiations per 15-Min Observation Sessions Among Study Participants: Stage I

Student	Baseline M	Intervention M	Maintenance M
Bob	0.09	0.67	0.00
Tevon	1.02	2.86	3.08
Josh	0.21	0.20	0.00

Table 3. Mean Numbers of Initiations per 15-Min Observation Sessions Among Study Participants: Follow-up Stage

Student	Baseline M	Intervention M	Maintenance M
Bob	0.05	0.00	0.00
Tevon	2.04	2.40	2.50
Josh	0.27	0.20	0.08

was such variation in his rate that it was unlikely that changes were related to VSM. There were no changes seen in the types of interactions he exhibited on the playground, which was in contrast to the generalized type of changes reported by Buggey et al. (2009). One noteworthy result with Tevon was that during baseline of the initial stage of the study, 38% of the observations resulted in no initiations being seen. During and after the intervention, he initiated in every observation session except the first one in the intervention phase. Similarly, in the second stage of the study, Tevon had zero initiations in 35% of the observations but initiated in all sessions thereafter.

Social Validity

Adults were solicited to provide information concerning the VSM procedure, how it was applied in the classrooms, the effects it had on students, and feasibility of these individuals using VSM in the future. Informal interviews were conducted beginning 1 week into the first intervention phase and continuing once every other week for the remainder of the study. Teachers, therapists, and parents were asked the same questions as in Buggey et al. (2009): "Do you believe VSM is helping the child?" "Does the implementation cause any disruption to the classroom routine?" "Does the child seem to [still] enjoy watching the videos?" "Is the implementation distracting to other students?" and (for parents) "Does your child communicate to you about being in a movie?" The procedure for making VSM videos was explained to the adults at the end of the study and they were asked about the practicality of implementing VSM on their own.

The three teachers seemed supportive of the method and agreed that the dependent variable was an area of concern and a skill important to the children's development. Two

reported almost no disruption to class routines and reported that the children enjoyed watching their videos. One teacher did feel that the method was too intrusive and took away time that could have been more productive for the child. The reports about treatment efficacy varied. Two teachers reported positive changes in behavior following the children viewing the videos while the same teacher who felt the method was intrusive saw no changes. Further questioning with the teachers who saw positive results indicated that they may have been hyper-vigilant or “hyper-hopeful,” paying closer attention to the children in terms of behavior change. In terms of practicality, the therapists and parents were more enthusiastic about actually making their own VSM videos. The teachers stated they had little time to devote to filming and editing and the training that would involve. The therapists were excited about adding this to their repertoires of interventions while parents expressed a willingness to learn more about the method.

Discussion

The implementation of a VSM intervention to facilitate social initiations on the playground appeared to be unsuccessful across three study participants, who were the youngest children ever to participate in such a study. Gains were not observed when VSM was reintroduced at a time when the children’s ages surpassed those in the earlier study by Buggey et al. (2009). Because the videos shown to the children in the follow-up stage of the study were identical to those shown earlier, it is possible that the children were desensitized to them resulting in decreased interest. Another factor that may have come into play is the enthusiasm levels exhibited by the children when first encountering their videos. Buggey et al. reported that all the children in their study watched the videos enthusiastically from the outset, often clapping and making comments. Two of the children in this study were hesitant to view the video at first.

Tevon’s results bear further consideration considering he was the child in this study whose age was closest to those in the earlier study. Tevon’s decrease in the number of sessions where he did not initiate (more than one third of baseline observations compared with zero following intervention) in both stages of the study is noteworthy. Besides being slightly older, Tevon differed from the other two children in that he had a baseline of initiations to work from. Josh and Bob were not seen initiating except in inappropriate ways such as hitting. Buggey et al. (2009) reported that in their study, the one child who did not make progress was limited to only one behavior throughout the study (physical approach to retrieve balls he rolled down slides). It may be

that at this young age some early skills in initiating social interactions are necessary for VSM to be effective.

It is also possible that the behavior selected for study was not age or ability appropriate. The idea of socially initiating may have been too far advanced developmentally. The behaviors were selected based on developmental appropriateness for typically developing children; however, it may be necessary to better evaluate antecedent skills for children on the autism spectrum. A focus on parallel play or proximity to other children might be more appropriate. That said, this was the identical behavior addressed by Buggey et al. (2009) where positive results were reported. It is also possible that initiations may have been too stressful for the children producing anxiety like that seen in older children with autism (Meyer, Mundy, van Hecke, & Durocher, 2006).

The results of this study, compared with those of Buggey et al. (2009), Bellini and Akullian (2007), and Wert and Neisworth (2003), present a quandary. This is the first study in which no results were evident for any participant. The critical factors seemed to be age and/or developmental levels, but there may be other variables affecting outcomes. As it stands now in the four studies, seven of eight 4-year-olds with autism made substantial progress when VSM was introduced and three children who were slightly less than 4 years old did not. Certainly more work needs to be done to determine the age factor associated with efficacy, whether there are individual child attributes that contribute to success or failure, and whether there are behaviors more amenable to change with VSM.

Limitations

A constant threat to validity in single-case designs is the small sample size. Even though the researchers attempted to select children who were similar in abilities with similar demographics, individual differences within such a small group of children may constitute a significant threat. Assessment instruments used in this study and in Buggey et al. (2009; for example, CARS and GARS) may not have been sensitive enough to pick up subtle differences in the children’s skills or the nature of their ASD.

Another threat that arose in this study dealt with treatment fidelity. Teacher initiation of the videos went unobserved and was monitored based on teacher reports. Because teachers showed the videos in their small offices off the classrooms (except for Josh who was reluctant to go into the teacher’s office), observation was impractical. Training the teachers on the laptop’s operation and making viewing the videos a one-click operation was done in the hope that the operation would be foolproof; however, this did not take into consideration that teachers might forget to show the video during the allotted times because of demands on their

time due to regular and serendipitous classroom activity. More careful control of treatment fidelity should be undertaken in future study.

Conclusions

The findings of the present study raise questions about the efficacy of VSM use on social initiations of 3-year-old children with an ASD. Studies using VSM with slightly older groups of preschoolers have been successful (i.e., Bellini & Akullian, 2007; Buggey et al. 2009; Wert & Neisworth, 2003) leading to the assumption that as we work with progressively younger children, we may experience variables that work against the success of VSM. Factors such as age, specific attributes of individuals, or a combination of factors may determine success or failure of the method with this age group. Behaviors targeted for intervention also must be carefully scrutinized to ensure that they are not only age-appropriate but also developmentally appropriate for the child. Certainly, much more research needs to be carried out to better understand the limitations of VSM and to determine prerequisite skills necessary for success.

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References

- Ayres, K. M., & Langone, J. (2005). Intervention and instruction with video for students with autism: A review of the literature. *Education and Training in Developmental Disabilities, 40*, 183–196.
- Bandura, A. (2001). Social cognitive theory: An agentic perspective. *Annual Review of Psychology, 52*, 1–26.
- Bayley, N. (2005). *Bayley Scales of Infant and Toddler Development* (3rd ed.). San Antonio, TX: Pearson Assessment.
- Bellini, S., & Akullian, J. (2007). A meta-analysis of video modeling and video self-modeling interventions for children and adolescents with autism spectrum disorders. *Exceptional Children, 73*, 261–284.
- Bellini, S., Akullian, J., & Hopf, A. (2007). Increasing social engagement in young children with autism spectrum disorders using video self-modeling. *School Psychology Review, 36*, 80–90.
- Buggey, T. (2005). Applications of video self-modeling with children with autism in a small private school. *Focus on Autism and Other Developmental Disabilities, 20*, 180–204.
- Buggey, T. (2007). A picture is worth: Video self-modeling applications at school and home. *Journal of Positive Behavior Interventions, 9*, 151–158.
- Buggey, T., Hoopes, G., Sherberger, M. E., & Williams, S. (2011). Facilitating social initiations of preschoolers with autism spectrum disorders using video self-modeling. *Focus on Autism and Other Developmental Disabilities, 26*, 25–36.
- Clark, E., Beck, D., Sloane, H., Goldsmith, D., Jensen, W., Bowen, J., & Kehle, T. (1993). Self-modeling with preschoolers: Is it different. *School Psychology International, 14*, 83–89.
- Delano, M. E. (2007). Video modeling interventions for individuals with autism. *Remedial and Special Education, 28*, 33–42.
- Dowrick, P. W. (1983). Self-modeling. In P. W. Dowrick & J. Biggs (Eds.), *Using video: Psychological and social applications* (pp. 105–124). New York, NY: John Wiley.
- Dowrick, P. W., & Raeburn, J. M. (1995). Self-modeling: Rapid skill training for children with physical disabilities. *Journal of Developmental and Physical Disabilities, 7*, 25–36.
- Gilliam, J. E. (2006). *Gilliam Autism Rating Scale* (2nd ed.). San Antonio, TX: Pearson Assessment.
- Harrison, P., & Oakland, T. (2003). *Adaptive Behavior Assessment System* (2nd ed.). San Antonio, TX: Pearson Assessment.
- Hitchcock, C. H., Dowrick, P. W., & Prater, M. A. (2003). Video self-modeling intervention in school-based settings: A review. *Remedial and Special Education, 24*, 36–45.
- Horner, R. H., Carr, E. G., Halle, J., McGee, G., Odom, A., & Wolery, M. (2005). The use of single-subject research to identify evidence-based practice in special education. *Exceptional Children, 71*, 165–179.
- Kennedy, C. (2005). *Single-case designs for educational research*. Boston, MA: Pearson.
- McCoy, K., & Hermansen, E. (2007). Video modeling for individuals with autism: A review of model types and effects. *Education and Treatment of Children, 30*, 183–213.
- Meyer, J. A., Mundy, P. C., van Hecke, A. V., & Durocher, J. S. (2006). Social attribution processes and comorbid psychiatric symptoms in children with Asperger syndrome. *Autism, 10*, 383–402.
- Schopler, E., Reichler, R. J., & Rochen Renner, B. (1988). *Childhood Autism Rating Scale*. Torrance, CA: Western Psychological Services.
- Wert, B. W., & Neisworth, J. T. (2003). Effects of video self-modeling on spontaneous requesting in children with autism. *Journal of Positive Behavior Interventions, 5*, 300–305.